Functions, Rules, Domain & Range

WARM UPS

TEACHER COPY | Display student pages; keep solutions private

1 | Solve

Solve for x. Show each step, then check your answer.

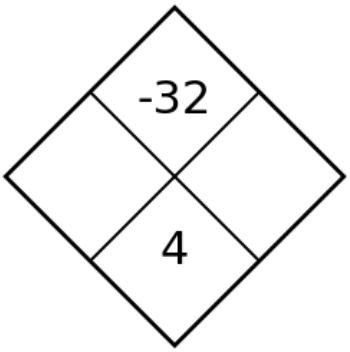
$$7x - 4 + 2x = 23$$

ANSWER / TEACHING NOTE

7x - 4 + 2x = 23 → 9x - 4 = 23 → 9x = 27 → x = 3.

2 | Diamond

Complete the diamond.



Functions, Rules, Domain & Range

WARM UPS

3 | Spiral Review

Evaluate $2a^2 - 3$ when $a = -2$. Show substitution.

ANSWER / TEACHING NOTE

2(-2)^2 - 3 = 2(4) - 3 = 5.

4 | Spiral Review

A pattern has outputs 5, 8, 11, 14 for inputs 0, 1, 2, 3. What is the change in output each step?

ANSWER / TEACHING NOTE

The output increases by 3 for each +1 input.

Functions, Rules, Domain & Range

LEARNING FOCUS & NOTES

Learning focus | LG 02-ican-1 / LG 02-ican-2 / LG 02-ican-3

- I can use an input-output rule to find outputs.
- I can explain why a relation is or is not a function.
- I can list the domain and range of a set of ordered pairs.

Vocabulary | Highlight the important words

Relation
A collection of input-output pairs; it may or may not be a function.

Function
A relation that assigns exactly one output to each input.

Input / output
An input is the starting value (often x); an output is the resulting value (often y).

Domain
The set of all distinct input values used by a relation.

Range
The set of all distinct output values produced by a relation.

Function rule
An equation or instruction that tells how to get an output from an input.

Meaning & Method | Rewritten review notes

1. Read an ordered pair as (input, output). A repeated input is allowed only if it gives the SAME output.

2. To test a function, inspect whether any input has two different outputs. On a graph, a vertical line should not hit more than one point.

3. For a finite relation, list each domain and range value only once; order from least to greatest.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

Functions, Rules, Domain & Range

EXAMPLES & YOU TRY ITS

Example

Is this relation a function? Explain why, then identify its domain and range.

Input x	Output y
-1	4
0	3
1	2
2	1

Decide if this relation is a function. List the domain and range.

> **ANSWER / TEACHING NOTE**
> Yes. Every input has only one output. Domain = {-1, 0, 1, 2}; range = {1, 2, 3, 4}.

You Try It

Determine whether this table represents a function. State the domain and range, and justify your answer.

Input x	Output y
-2	1
0	3
0	5
3	7

Function? Explain. Domain? Range?

> **Function? Explain. Domain? Range?**
> Not a function: the input 0 has outputs 3 and 5. Domain = {-2, 0, 3}; range = {1, 3, 5, 7}.

Why does the repeated input in this table matter?

> **TEACHING NOTE**
> Input 0 has two distinct outputs, 3 and 5, so the table is not a function. A duplicate pair with the same output would be allowed.

Functions, Rules, Domain & Range

LAUNCH

Function detective: four cases

Work with a partner. For each case, decide whether the relation is a function, then record its domain and range.

Case A

x	y
1	2
2	4
3	6

Case B

x	y
1	5
1	7
2	9

Case C

x	y
0	4
1	4
2	4

Case D

x	y
1	2
1	2

1 | Compare Cases B and D. Both repeat input 1. Why does only one fail the function test?

> **ANSWER / TEACHING NOTE**
> A, C, and D are functions; B is not. In B, input 1 has two different outputs (5 and 7). In D, input 1 repeats with the same output (2), so the relation remains a function. D has domain {1} and range {2}. A has domain {1,2,3} and range {2,4,6}; C has domain {0,1,2} and range {4}.

2 | Choose one function and predict another input-output pair from a pattern you observe. Is your new pair certain or a reasonable prediction? Explain.

> **ANSWER / TEACHING NOTE**
> Example for A: (4,8) follows a possible pattern, but the given table does not prove it must continue.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Functions, Rules, Domain & Range

REVIEW & PREVIEW

1 A function machine follows $y = 2x + 5$. Complete the outputs for inputs 0, 2, 4, and 6.

ANSWER / TEACHING NOTE

Outputs: 5, 9, 13, 17.

2 ★ For the relation {(-1,0), (0,2), (-1,3), (2,8)}, decide whether it is a function. Explain and list domain/range.

ANSWER / TEACHING NOTE

Not a function: -1 maps to 0 and 3. Domain {-1, 0, 2}; range {0, 2, 3, 8}.

3 Write two different ordered pairs with the same OUTPUT but different INPUTS. Does this break the function rule?

ANSWER / TEACHING NOTE

Example: (1,4) and (2,4). Yes, this can be a function; outputs may repeat.

Functions, Rules, Domain & Range

REVIEW & PREVIEW

4 ★ A game costs $5 to enter, plus $3 per round. Use inputs 0, 1, 2, 3 rounds to list the outputs and the domain/range.

> **ANSWER / TEACHING NOTE**
> Outputs: 5, 8, 11, 14. Domain {0, 1, 2, 3}; range {5, 8, 11, 14}.

5 Spiral: Solve $2(x+5)=24$. Show your steps.

> **ANSWER / TEACHING NOTE**
> 2(x+5)=24 → x+5=12 → x=7.

6 Preview: Table outputs are 9, 6, 3, 0 for inputs 1, 2, 3, 4. What stays constant as the input increases by 1?

> **ANSWER / TEACHING NOTE**
> Constant difference is -3 for each increase of 1 input.

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.

Tables, Graphs & Constant Change

WARM UPS

TEACHER COPY | Display student pages; keep solutions private

1 | Solve

Solve for x. Show each step, then check your answer.

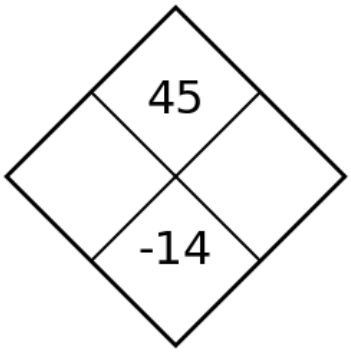
$$2(3x + 1) - 4 = 22$$

ANSWER / TEACHING NOTE

2(3x + 1) - 4 = 22 → 6x + 2 - 4 = 22 → 6x = 24 → x = 4.

2 | Diamond

Complete the diamond.



Tables, Graphs & Constant Change

WARM UPS

3 | Spiral Review

Is {(-1,4), (0,4), (2,4)} a function? Why?

> **ANSWER / TEACHING NOTE**
> Yes: each input has exactly one output; repeated outputs are allowed.

4 | Spiral Review

List the domain and range of {(2,7), (4,9), (6,7)}.

> **ANSWER / TEACHING NOTE**
> Domain {2,4,6}; range {7,9}.

Tables, Graphs & Constant Change

LEARNING FOCUS & NOTES

Learning focus | LG 02-ican-4 / LG 03-ican-1

- I can recognize constant additive change in a table.
- I can compare vertical change with horizontal change.
- I can connect a table, graph, and equation that describe the same linear pattern.

Vocabulary | Highlight the important words

Linear relationship

A relationship that has a constant rate of change and graphs as a straight line.

Constant difference

The same change in output for each equal change in input.

Horizontal change (Δx)

The difference between x-coordinates of two points; movement left or right.

Vertical change (Δy)

The difference between y-coordinates of two points; movement up or down.

Slope

The ratio of vertical change to horizontal change, Δy / Δx.

Starting value

The output at input zero; where a linear graph crosses the y-axis.

Meaning & Method | Rewritten review notes

1. Compare equal input intervals when checking for a linear pattern. A constant output difference is meaningful only when the input step is constant.

2. Compute slope as change in y divided by change in x. Keep the SAME point order in numerator and denominator.

3. For a linear rule y = mx + b, m is the slope and b is the output when x = 0.

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Tables, Graphs & Constant Change

EXAMPLES & YOU TRY ITS

Example

Calculate Δx and Δy from the table, find the slope, and write the equation of the line.

x	y
0	2
2	8
4	14

Find Δx and Δy. Calculate the slope and write the linear equation.

> **ANSWER / TEACHING NOTE**
> Δx = +2, Δy = +6; m = 6/2 = 3; initial output b = 2.
> Equation y = 3x + 2.

You Try It

Use this table to find Δx and Δy, calculate the slope, and write a linear equation.

x	y
0	11
2	7
4	3

Find Δx, Δy, slope, and equation.

> **Show Δx, Δy, slope, and equation.**
> Δx = 2; Δy = -4; m = -4/2 = -2; initial output b = 11.
> Equation y = -2x + 11.

How can you verify that the rate of change stays constant?

> **TEACHING NOTE**
> Every +2 change in x has a -4 change in y, so the slope stays -2 throughout.

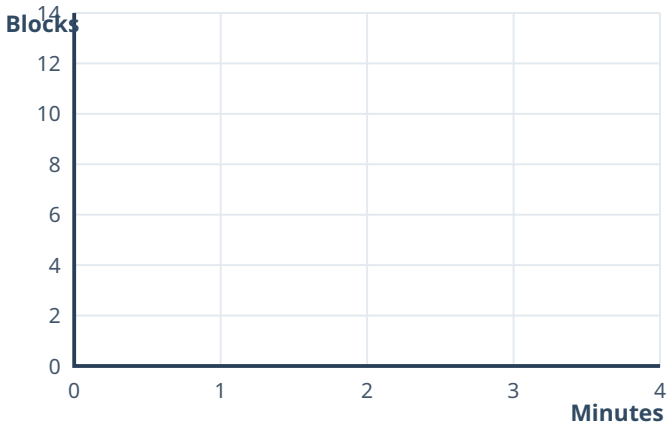
Tables, Graphs & Constant Change

LAUNCH

One story, three views

A walker starts 2 blocks from school and moves 3 more blocks away each minute. Complete the table and sketch the points; connect them if time is continuous.

Time (min)	Blocks from school
0	2
1	___
2	___
3	___



1 | How is the +3 visible in the table and on the graph? Write a rule using distance d and time t.

> **ANSWER / TEACHING NOTE**
> Table outputs: 2, 5, 8, 11. Graph these points. Rise = 3 for run = 1. Equation d = 3t + 2.

2 | A different walker starts 5 blocks away and moves 2 blocks each minute. Compare their changes for one minute (do not solve where they meet).

> **ANSWER / TEACHING NOTE**
> Walker 1 rate 3 blocks/min versus Walker 2 rate 2 blocks/min.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Tables, Graphs & Constant Change

REVIEW & PREVIEW

1 Find the slope of a table with x-values 0,2,4,6 and y-values 6,12,18,24. Explain each change.

ANSWER / TEACHING NOTE

Δy = +6 for Δx = +2, so slope = 3.

2 A line passes through (0,6), (2,10), and (4,14). Find its slope and starting value.

ANSWER / TEACHING NOTE

Slope = (10 - 6)/(2 - 0) = 2; starting value = 6.

3 ★ The table (0,8), (1,5), (2,2), (3,-1) is linear. Find slope and write the rule.

ANSWER / TEACHING NOTE

m = -3 and b = 8, so y = -3x + 8.

Tables, Graphs & Constant Change

REVIEW & PREVIEW

4 ★ Compare: Pattern A adds 2 to the output for each +1 input. Pattern B adds 4. Which graph is steeper? Explain.

ANSWER / TEACHING NOTE

B is steeper: its positive slope 4 is greater than A's positive slope 2.

5 Spiral: Is {(1,2),(1,2),(3,5)} a function? State the domain and range.

ANSWER / TEACHING NOTE

Yes; repeating exactly (1,2) does not violate the function rule. Domain {1,3}; range {2,5}.

6 Preview: A runner gains 18 meters in 3 seconds. What is the average rate of change, including units?

ANSWER / TEACHING NOTE

18 m / 3 s = 6 m/s.

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.

Slope, Motion & Comparing Rates

WARM UPS

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1 | Solve

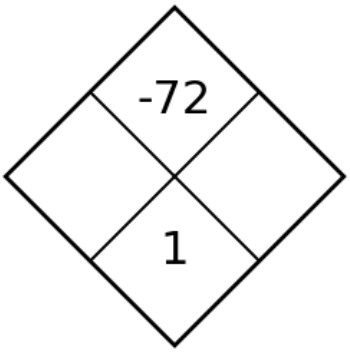
Solve for x. Show each step, then check your answer.

$$5(x-2)-2x=14$$

> **ANSWER / TEACHING NOTE**
> 5(x - 2) - 2x = 14 → 5x - 10 - 2x = 14 → 3x = 24 → x = 8.

2 | Diamond

Complete the diamond.



Slope, Motion & Comparing Rates

WARM UPS

3 | Spiral Review

A table includes (0,5), (2,11), (4,17). Find its slope.

ANSWER / TEACHING NOTE

m = (11 - 5)/(2 - 0) = 6/2 = 3.

4 | Spiral Review

Can {(1,8), (2,8), (2,9)} be a function? Explain.

ANSWER / TEACHING NOTE

No: input 2 produces two different outputs (8 and 9).

Slope, Motion & Comparing Rates

LEARNING FOCUS & NOTES

Learning focus | LG 02-ican-5 / LG 03-ican-3 / LG 03-ican-4

- I can calculate slope from two points or a table.
- I can interpret a slope using units and a direction.
- I can compare two rates fairly, including negative slopes.

Vocabulary | Highlight the important words

Rate of change

How much an output changes compared with an input change, with units attached.

Slope

The vertical change divided by the horizontal change of a straight line.

Positive / negative slope

A positive slope rises as x increases; a negative slope falls as x increases.

Zero slope

A horizontal line with no vertical change; slope zero.

Position

An object's location measured relative to a chosen reference point.

Velocity on a position-time graph

The rate and direction of change in position; the slope is signed and measured in distance per time.

Meaning & Method | Rewritten review notes

1. Always find slope with a ratio: (change in output) / (change in input), not the vertical change alone.

2. In context, read the sign and the units. A negative position-time slope can mean travel toward the chosen zero position; it does not mean negative speed.

3. To compare steepness or speed, compare magnitudes when the units are the same; explain the signs separately as direction or decreasing quantity.

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Slope, Motion & Comparing Rates

EXAMPLES & YOU TRY ITS

Example

Find the cart's slope in meters per second. Explain how the sign describes its motion.

Time (s)	Position (m)
0	30
2	22
4	14

Calculate the slope in meters per second. Explain the sign and direction.

> **ANSWER / TEACHING NOTE**
> Δposition = 22 - 30 = -8 m; Δtime = 2 s; slope = -8/2 = -4 m/s. Position decreases by 4 m each second.

You Try It

Use this position-time table to calculate the cart's slope in m/s and explain its direction of motion.

Time (s)	Position (m)
0	10
3	25
6	40

Find slope with units and explain direction.

> **Find slope with units. What does its sign tell you?**
> Δposition = 25 - 10 = +15 m; Δtime = 3 s; slope = +5 m/s; position increases 5 m each second.

What does a positive or negative slope say about motion?

> **TEACHING NOTE**
> Positive slope means position increases over time; negative slope means position decreases relative to the reference direction.

Slope, Motion & Comparing Rates

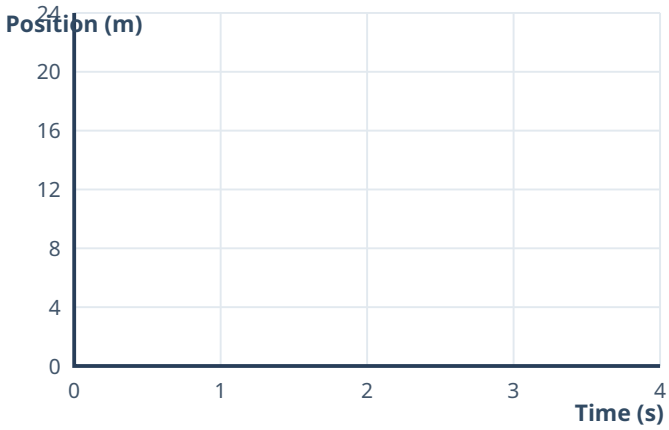
LAUNCH

Two walkers, one coordinate system

The values are positions measured in meters from the same starting marker. Graph both walkers using the given data, and explain how slope indicates direction.

Time (s)	Walker A (m)	Walker B (m)
0	0	20
1	4	17
2	8	14
3	12	11

Graph key: Walker A is blue; Walker B is red.



1 | Calculate each signed slope. Which walker changes position faster in magnitude? How do you know?

> **ANSWER / TEACHING NOTE**
> Walker A: +4 m/s. Walker B: -3 m/s. Walker A has greater speed magnitude, 4 m/s.

2 | What does the negative slope for Walker B mean? Why would it be inaccurate to call Walker B's speed -3 m/s?

> **ANSWER / TEACHING NOTE**
> B moves toward smaller coordinate values; its velocity is -3 m/s but speed is 3 m/s.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Slope, Motion & Comparing Rates

REVIEW & PREVIEW

1 ★ Position goes from 32 m at 0 s to 14 m at 6 s. Calculate average rate of change with units.

ANSWER / TEACHING NOTE

(14 - 32)/(6 - 0) = -18/6 = -3 m/s.

2 Find the slope between (-2,7) and (4,-5). Show Δy and Δx.

ANSWER / TEACHING NOTE

(-5 - 7)/(4 - (-2)) = -12/6 = -2.

3 ★ Two cyclists travel 30 m in 5 s and 42 m in 7 s. Compare their average rates.

ANSWER / TEACHING NOTE

Both are 6 m/s: 30/5 = 6 and 42/7 = 6.

Slope, Motion & Comparing Rates

REVIEW & PREVIEW

4 A tank starts at 70 gallons and loses 4 gallons per minute. Interpret the starting value and rate.

ANSWER / TEACHING NOTE

Initial volume 70 gallons; change -4 gallons each minute.

5 Spiral: For {(0,5),(2,5),(2,7)}, is it a function? List domain/range.

ANSWER / TEACHING NOTE

Not a function: x=2 has two outputs. Domain {0,2}; range {5,7}.

6 Preview: A scooter is 12 meters from a marker at time 0 and its position increases by 3 meters each second. Write an equation for position p after t seconds.

ANSWER / TEACHING NOTE

p = 3t + 12 (meters, seconds).

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.

Writing & Interpreting Linear Models

WARM UPS

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1 | Solve

Solve for x. Show each step, then check your answer.

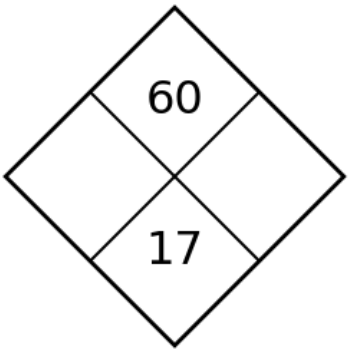
$$2(3x - 4) + 5 = 27$$

ANSWER / TEACHING NOTE

2(3x - 4) + 5 = 27 → 6x - 8 + 5 = 27 → 6x = 30 → x = 5.

2 | Diamond

Complete the diamond.



Writing & Interpreting Linear Models

WARM UPS

3 | Spiral Review

Calculate slope between (1,7) and (3,15).

> **ANSWER / TEACHING NOTE**
> m = (15 - 7)/(3 - 1) = 8/2 = 4.

4 | Spiral Review

Is {(0,2),(1,4),(2,6)} a function? List domain and range.

> **ANSWER / TEACHING NOTE**
> Yes. Domain {0,1,2}; range {2,4,6}.

Writing & Interpreting Linear Models

LEARNING FOCUS & NOTES

Learning focus | LG 03-ican-2 / LG 03-ican-5

- I can write y = mx + b from a description, table, or two points.
- I can explain what the slope and starting value mean.
- I can check whether a model makes sense for a stated input.

Vocabulary | Highlight the important words

Linear model

An equation, table, or graph using constant change to represent a situation.

Slope-intercept form

The equation y = mx + b, where m is slope and b is the output at x = 0.

Slope / rate m

The change in output for each 1-unit input increase; include units.

y-intercept / starting value b

The value of y when x = 0; often the starting amount in context.

Variable

A letter representing a changing or unknown quantity, with an assigned meaning.

Domain restriction

A limit on which inputs make sense, based on the situation or mathematical rule.

Meaning & Method | Rewritten review notes

1. Read the story first: choose an input x and an output y with meaningful units.

2. Identify the starting amount b at input zero, then identify the constant change m per one input unit. Write y = mx + b.

3. Check the model using an input and output from the context. State restrictions such as nonnegative whole-number counts.

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Writing & Interpreting Linear Models

EXAMPLES & YOU TRY ITS

Example

A printing service charges a $12 setup fee plus $3 per poster. Write the cost equation, explain the fee and per-poster rate, and find the cost for 5 posters.

Posters p	Cost C ($)
0	12
1	15
2	18

Write the cost equation. Explain the fees and find the cost of 5 posters.

> **ANSWER / TEACHING NOTE**
>
> b = $12 setup fee, m = $3/poster. C = 3p + 12; for p=5, C = $27. Count p is a nonnegative integer.

You Try It

A studio charges a $9 reservation fee plus $4 per hour. Complete the table, write the cost equation, explain the slope and starting value, and calculate the cost for 3 hours.

Hours h	Cost C ($)
0	___
1	___
2	___

Write the equation, interpret m and b, find cost for 3 hours.

> **Write the equation, interpret the $9 and $4, and find cost for 3 hours.**
>
> C = 4h + 9; b = $9 reservation fee; m = $4/hour; cost for 3 hours is $21.

How do the setup fee and rate appear in your equation?

> **TEACHING NOTE**
>
> The starting fee is the value at zero (intercept); the hourly or per-item charge is the slope.

Writing & Interpreting Linear Models

LAUNCH

Which rental plan fits?

Two bike rental plans are offered. Plan A costs $6 plus $4 per hour. Plan B costs $10 plus $3 per hour. Compare their costs at the times shown in the table.

Hours h	Plan A ($)	Plan B ($)
0	___	___
1	___	___
2	___	___
3	___	___

Plan A: $A = 4h + 6$

Plan B: $B = 3h + 10$

Use the table to make your comparison for these specific times.

1 | What does each plan's slope mean? What does each starting value mean?

ANSWER / TEACHING NOTE

Hours 0,1,2,3: Plan A costs 6,10,14,18; Plan B costs 10,13,16,19.

2 | Which is cheaper at 1 hour? 3 hours? Explain using numbers and words.

ANSWER / TEACHING NOTE

Slope A=$4/hr, start $6; slope B=$3/hr, start $10. Plan A is cheaper at 1 hour and 3 hours.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Writing & Interpreting Linear Models

REVIEW & PREVIEW

1 ★ A membership costs $5 per month plus $2 for each movie rented. Write a model for total monthly cost C for n movies. Interpret m and b.

ANSWER / TEACHING NOTE

C = 2n + 5. Rate $2/movie and starting fee $5/month.

2 The table has points (0,15), (1,13), (2,11), (3,9). Write y = mx + b and interpret its trend.

ANSWER / TEACHING NOTE

m = -2, b = 15; y = -2x + 15.

3 ★ Write the equation of a line through (0,4) and (2,10). Show work for m and b.

ANSWER / TEACHING NOTE

m = (10 - 4)/2 = 3; b = 4; y = 3x + 4.

Writing & Interpreting Linear Models

REVIEW & PREVIEW

4 Choose which rule matches outputs 4, 7, 10 at inputs 0, 1, 2: (a) y=3x+4 (b) y=4x+3 (c) y=7x+4. Explain.

ANSWER / TEACHING NOTE

Choice (a): y = 3x + 4.

5 Position is modeled by p = 20 - 5t (meters, seconds). Interpret 20 and -5; predict position at t = 3.

ANSWER / TEACHING NOTE

Initial position 20 m; velocity -5 m/s; p(3) = 5 m.

6 Spiral + model check: Is {(0,7),(1,9),(2,11),(2,11)} a function? What linear rule fits these pairs?

ANSWER / TEACHING NOTE

Yes: the repeated pair is identical. Rule y = 2x + 7.

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.